

MECH 422 – Advanced Materials & Processes

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Failure Analysis of HYTORC 360° Rotatable Reaction Arm

Group 3

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Project *overview*

1

Defective Part &
Failure

2

Material &
Identification

3

Processing
History

4

Recycling
Feasibility

5

Alternative Materials
& Processes

01 DEFECTIVE PART & FAILURE

HYTORC 360° Rotatable Reaction Arm

What It Does

The reaction arm provides support during torque applications

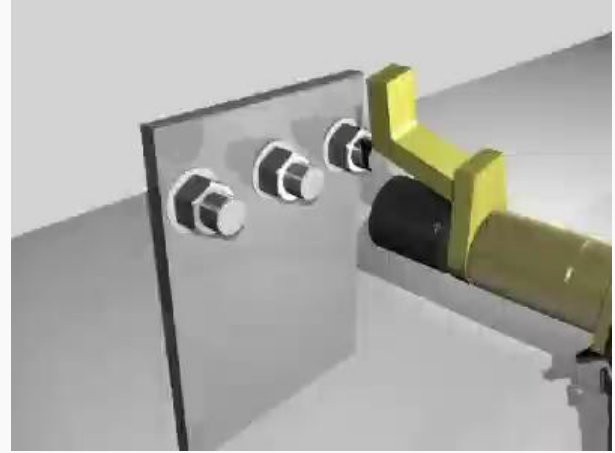
- Swivel flap connects to torque wrench via gear teeth
- Transfers reaction forces to base component
- Prevents tool movement
- 360° rotation for confined spaces

Formula Club Use

Likely used for assembly of high-strength bolt joints in:

- Suspension connections
- Gearbox assemblies
- Chassis connections

This component is assumed to be manufactured from a **hardened steel** designed for high strength and loading.



HYTORC®

01 DEFECTIVE PART & FAILURE

Failure Characteristics

Failure Location

Complete fracture at base connection where swivel flap meets reaction arm base

Probable Cause

Excessive torque combined with insufficient support during operation

Material Behaviour

Brittle fracture consistent with hardened steel properties



Crack Initiation

Dense lines and sharp surface features mark the origin point

Crack Propagation

Circular bands radiating from initiation site indicate progressive failure

Rapid Fracture

Smoother texture characteristic of brittle, catastrophic failure

O2 MATERIAL & IDENTIFICATION

Material Identification Process

Initial assumption: hardened carbon steel. After comprehensive laboratory testing, we were able to make a much more accurate hypothesis.

01

Density Testing

Archimedes' Principle method measured density at 7.69 g/cm^3 , consistent with alloy steel composition.

03

SEM Imaging

Revealed rounded inclusions, crack deflection patterns, and mixed-mode failure with dimples and cleavage facets.

02

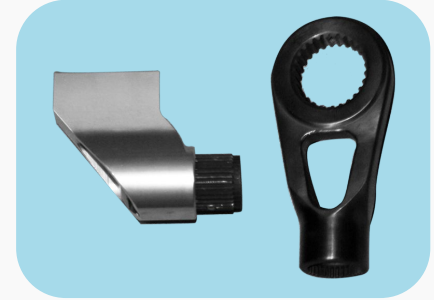
Hardness Analysis

Vickers test yielded 404.6 HV. Mohs test provided 4, corresponding to 150-200 HV range for validation.

04

Strength Calculation

Hall-Petch equation estimated 211.8 MPa baseline, with heat treatment increasing to 600-1400 MPa range.

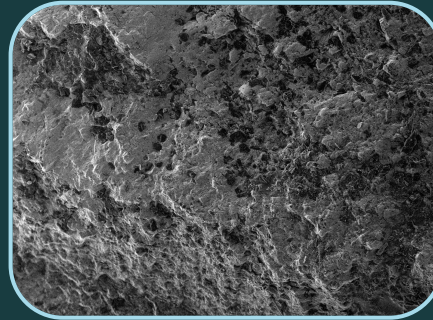
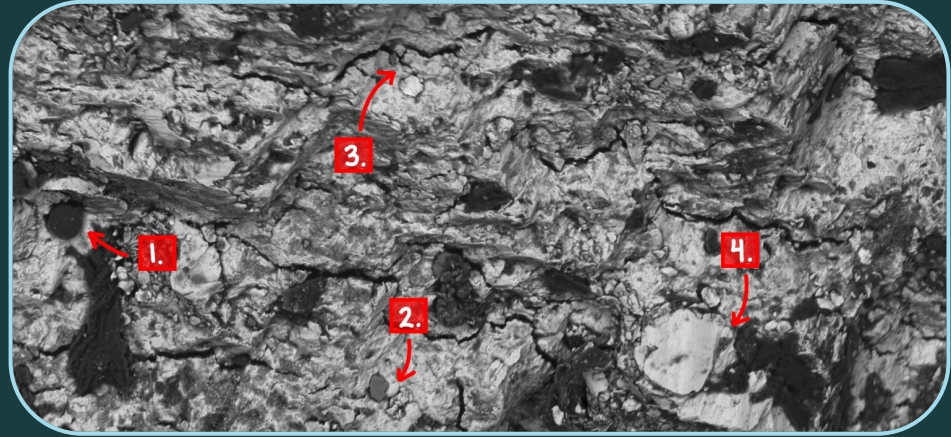


O2 MATERIAL & IDENTIFICATION

Microstructure Evidence

Key Fracture Features

1. **Rounded inclusions** embedded in matrix indicating alloy composition
2. **Transgranular fracture** with crack deflection around particles
3. **Secondary cracking** perpendicular to main crack from cyclic loading
4. **Cleavage facets** showing mixed-mode failure
5. **Ductile dimples** from grain pull-out under repeated deformation

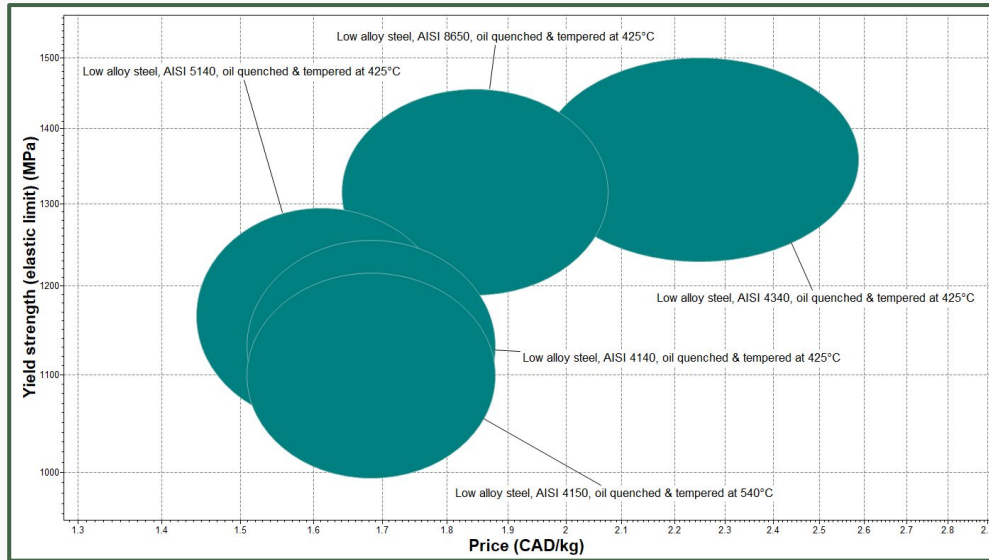


Material Information

- Alloy steel
- Material property values
- Failure mode
- Application

O3 PROCESSING HISTORY

Material Selection Rationale



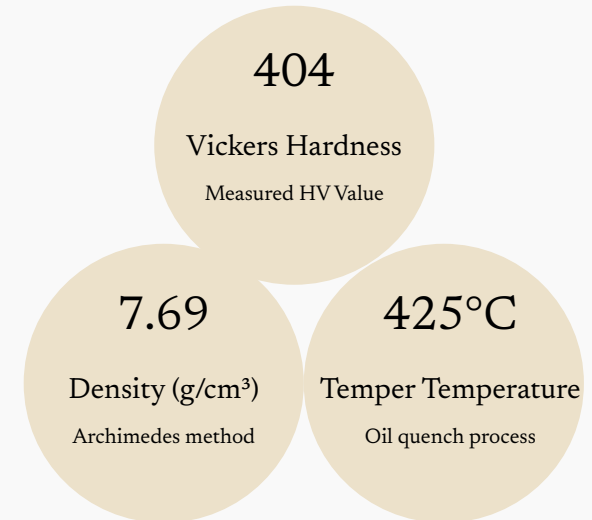
Connection to Working Principle

The reaction arm transfers counter-torque from hydraulic/pneumatic wrenches to fixed structures, requiring resistance to **alternating torsional and bending stresses**.

Why AISI 4140?

AISI 4140 chromium-molybdenum steel provides optimal balance of strength, toughness, and hardenability at moderate cost for torque-reaction tooling components.

- Chromium: Improves **hardenability**
- Molybdenum: Reduces embrittlement during **cyclic loading**
- Cost-effective: More **economical** than alternatives



O3 PROCESSING HISTORY

Manufacturing Process

AISI 4140



Blast Furnace

Iron ore, coke, and limestone produce molten pig iron with hot air blasting from bottom.



Basic Oxygen Furnace

Oxygen blasted into pig iron reduces carbon content. Chromium and Molybdenum alloys added for AISI 4140.



Casting

Refined steel poured into molds to create ingots or billets for further processing.

Heat Treatment Process

Annealing

Heated and slowly cooled to reduce internal stresses

1

2

Pre-Heating

Heated below austenitization phase to avoid thermal shock.

Austenitization

Ferrite transforms to austenite

3

4

Oil Quenching

Rapid cooling at ~425°F transforms austenite to desired martensite structure.

5

Tempering

Heating and slow cooling prevents martensite cracking.

O4 RECYCLING FEASIBILITY

Material Separation Process

01

Secure and Remove Fasteners

Place part in vice, undo set screw, and remove straight pin

03

Separate Components

Remove springs, detent balls, and end plate from shaft assembly

02

Disassemble Shafts

Remove stainless steel cap and middle/inner shafts from main body

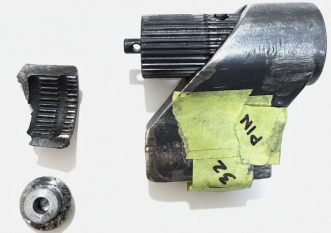
04

Clean & Sort

Degrease parts and separate low-alloy steel into ferrous bin, stainless into stainless bin

**Disassembly Time**

Complete separation takes **10–15 minutes**. All components are steel-based with no bonded interfaces, making material separation highly feasible.



O4 RECYCLING FEASIBILITY

Recycling Pathways & Environmental Impact



Low-Alloy Steel

Bulk remelting in Electric Arc Furnace (EAF) using ~95% recycled content. Size reduction, magnetic separation, then melting and casting.



Stainless Steel

Similar EAF process plus Vacuum-Oxygen Decarburization (VOD). Undamaged components can be remanufactured for reuse.



Eco-Audit Results

~4.95 kg CO₂

Recycled:

Low-Alloy Steel + Stainless Steel

Total emissions (8.43kg - 3.46kg
end-of-life credit)

Recycled: Low-Alloy Steel

Remanufacture: Stainless Steel

Total emissions (8.42kg - 3.47kg
end-of-life credit)

Steel is 100% recyclable and the most commonly recycled material in North America. Both recycling and remanufacturing options show negative end-of-life potential, offsetting total emissions significantly.

O5 ALTERNATIVE MATERIALS & PROCESSES

Alternative Materials for Improved Performance

High-Carbon Steel AISI 1080

Pros:

- Higher strength
- No process changes

Cons:

- More brittle
- Worse impact tolerance

Metal Matrix Composite Al 2024-B(f)

Pros:

- Very high fatigue strength
- Lightweight

Cons:

- Very expensive
- Hard to form complex shapes

Ceramic Matrix Composite WC-Co

Pros:

- Extremely high fatigue strength
- Excellent wear resistance

Cons:

- Brittle
- Not castable for this geometry

Polymer Matrix Composite E-glass Fiber

Pros:

- High strength-to-weight
- Good insulation

Cons:

- Low stiffness
- Not ideal for load-bearing parts

Advanced Alloys Ti-6Al-4V/Inconel 718

Pros:

- Excellent fatigue strength
- High corrosion resistance

Cons:

- Very high cost
- Requires specialized processing

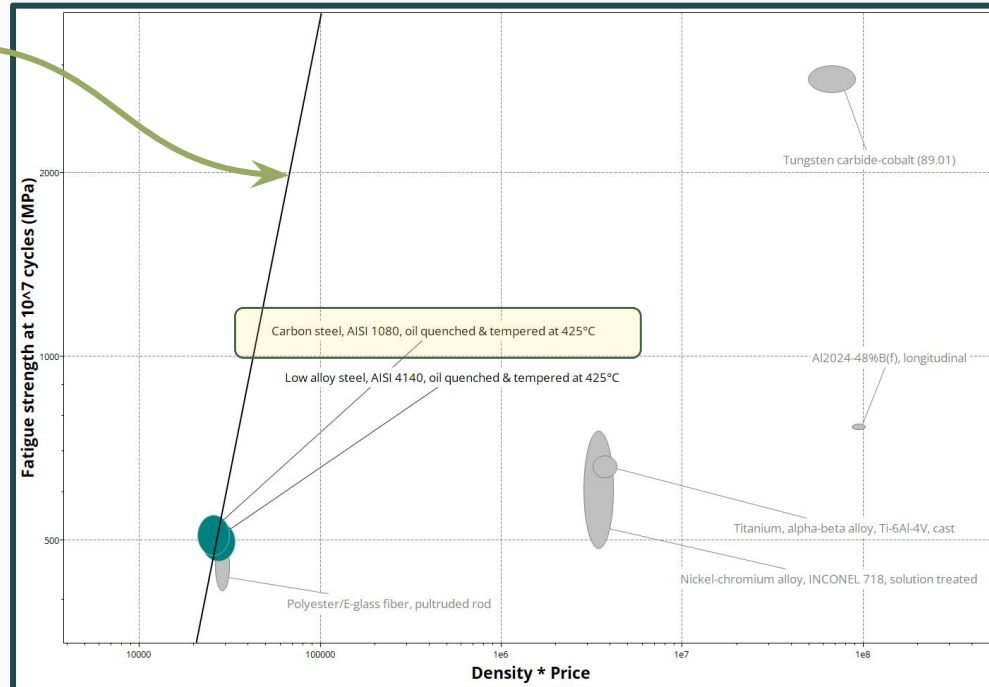
O5 ALTERNATIVE MATERIALS & PROCESSES

Granta Comparison Chart:

Fatigue Strength vs. Density*Price for Alternative Materials

**Material
Performance
Index (MPI)**

$$MPI = \frac{C_m \cdot \rho}{\sigma_e^{2/3}}$$



O5 ALTERNATIVE MATERIALS & PROCESSES

Alternative Processing Routes

Induction Hardening

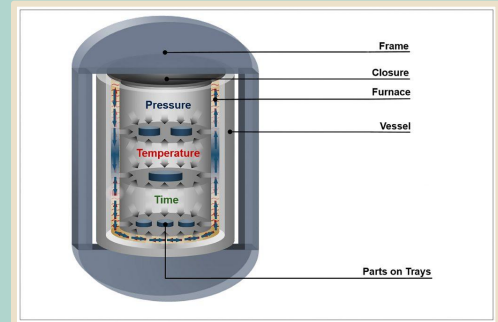
Induction hardening rapidly heats and quenches only the surface, creating a hard outer layer while keeping a tough core. This is a good option because it will **raise the reaction arm's surface fatigue strength** without changing the casting process.

Nitriding

Nitriding diffuses nitrogen into the steel at low temperature to form a hard, wear-resistant surface. It is suitable for our part because it **boosts surface hardness and fatigue resistance** with virtually no distortion.

Hot Isostatic Pressing (HIP)

HIP applies high heat and pressure to close internal pores in castings. Since our failed arm showed casting defects, HIP is a strong candidate for **improving durability** by removing voids that initiate fatigue cracks.



O5 ALTERNATIVE MATERIALS & PROCESSES

Recyclability of the Alternative Design

Current Material
Low-Alloy Steel (4140)

4.96

Recycled Option

kg CO₂ total emissions (8.43kg -
3.46kg end-of-life credit)

Alternative Material
High-Carbon Steel (1080)

4.99

Recycled Option

kg CO₂ total emissions (8.46kg -
3.47kg end-of-life credit)

-
- **Nearly identical recyclability** and Eco Audit values
 - Slightly more alloying, but still fully recyclable in standard BOF/EAF routes

- Can be fully recycled with minimal property loss
- Fits into Canada's >90% scrap EAF steelmaking stream
- **Simpler chemistry than 4140 → easier scrap integration**

Thank you

Q&A

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